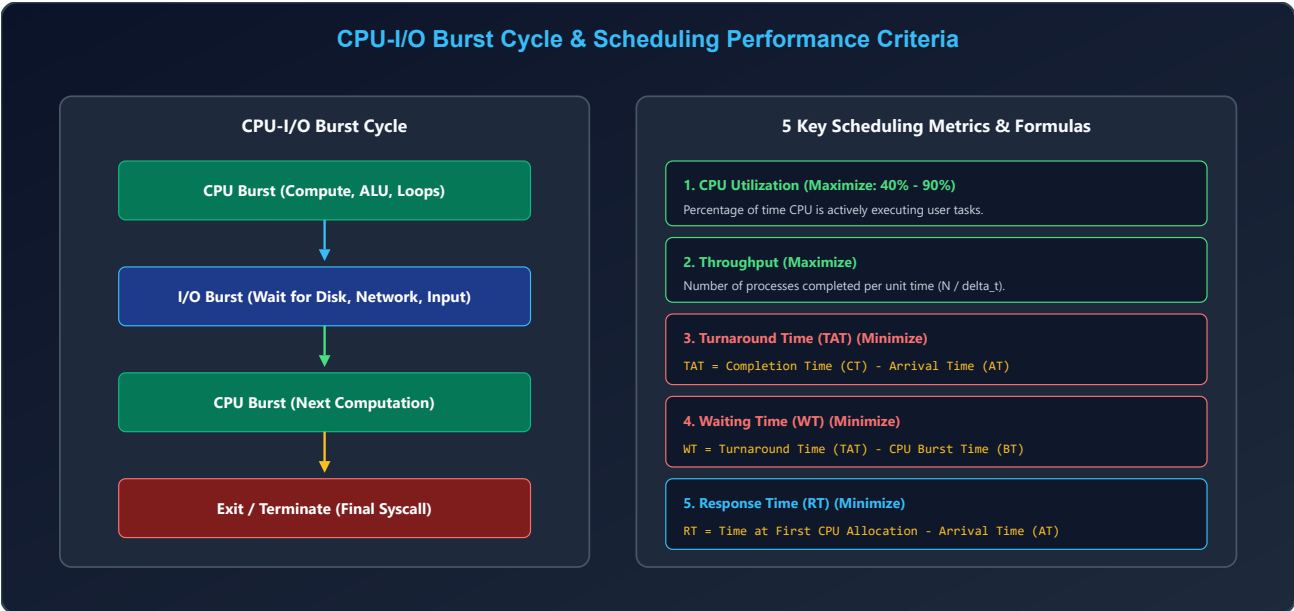


Module 04: CPU Scheduling Concepts & Performance Criteria

1. CPU-I/O Burst Cycle & Scheduling Metrics



2. Golden Mathematical Formulas

Metric Name	Mathematical Formula	Optimization Objective
Turnaround Time (TAT)	$TAT = Completion\ Time\ (CT) - Arrival\ Time\ (AT)$	MINIMIZE
Waiting Time (WT)	$WT = Turnaround\ Time\ (TAT) - Burst\ Time\ (BT)$	MINIMIZE
Response Time (RT)	$RT = First\ Response\ Time - Arrival\ Time\ (AT)$	MINIMIZE
Throughput	$Completed\ Processes / Total\ Time$	MAXIMIZE
CPU Utilization	$(Active\ CPU\ Time / Total\ Time) * 100\%$	MAXIMIZE

3. Preemptive vs Non-Preemptive Summary

Non-Preemptive: Process willingly CPU release karta hai (I/O or Terminate). $RT == WT$.

Preemptive: OS timer interrupt ya higher priority aane par forcibly CPU preempt kar sakta hai.